



Aggregating Data Using Group Functions





Objectives

- After completing this lesson, you should be able to do the following:
 - Identify the available group functions
 - Describe the use of group functions
 - Group data using the GROUP BY clause
 - Include or exclude grouped rows by using the HAVING clause



What Are Group Functions?

- Group functions operate on sets of rows to give one result per group.

EMP

DEPTNO	SAL
10	2450
10	5000
10	1300
20	800
20	1100
20	3000
20	3000
20	2975
30	1600
30	2850
30	1250
30	950
30	1500
30	1250

“maximum salary in the EMP table”

MAX (SAL)
5000



Types of Group Functions

- AVG
- COUNT
- MAX
- MIN
- SUM





Using Group Functions

```
SELECT [column,] group_function(column)
FROM      table
[WHERE condition]
[GROUP BY column]
[ORDER BY column];
```





Using AVG and SUM Functions

- You can use AVG and SUM for numeric data.

```
SQL> SELECT  AVG(sal), MAX(sal),  
2      MIN(sal), SUM(sal)  
3 FROM      emp  
4 WHERE     job LIKE 'SALES%';
```

AVG (SAL)	MAX (SAL)	MIN (SAL)	SUM (SAL)	
-----	-----	-----	-----	
1400	1600	1250	5600	





Using MIN and MAX Functions

- You can use MIN and MAX for any datatype.

```
SQL> SELECT MIN(hiredate), MAX(hiredate)
2 FROM emp;
```

MIN (HIRED	MAX (HIRED
-----	-----
17-DEC-80	12-JAN-83





Using the COUNT Function

- COUNT(*) returns the number of rows in a table.

```
SQL> SELECT COUNT (*)  
2 FROM emp  
3 WHERE deptno = 30;
```

COUNT (*)
6





Using the COUNT Function

- COUNT(*expr*) returns the number of nonnull rows.

```
SQL> SELECT COUNT(comm)
      2 FROM emp
      3 WHERE deptno = 30;
```

```
COUNT (COMM)
```

```
-----
```

```
4
```





Group Functions and Null Values

- Group functions ignore null values in the column.

```
SQL> SELECT AVG(comm)
      2 FROM emp;
```

AVG (COMM)

550



Creating Groups of Data

EMP

DEPTNO	SAL
10	2450
10	5000
10	1300
20	800
20	1100
20	3000
20	3000
20	2975
30	1600
30	2850
30	1250
30	950
30	1500
30	1250

2916.6667

2175

1566.6667

“average
salary
in EMP
table
for each
department”

DEPTNO	AVG (SAL)
10	2916.6667
20	2175
30	1566.6667



Creating Groups of Data: GROUP BY Clause

```
SELECT column, group_function(column)
FROM      table
[WHERE condition]
[GROUP BY group_by_expression]
[ORDER BY column];
```

- Divide rows in a table into smaller groups by using the GROUP BY clause.





Using the GROUP BY Clause

- All columns in the SELECT list that are not in group functions must be in the GROUP BY clause.

```
SQL> SELECT deptno, AVG(sal)
2 FROM emp
3 GROUP BY deptno;
```

DEPTNO	AVG (SAL)
10	2916.6667
20	2175
30	1566.6667





Using the GROUP BY Clause

- The GROUP BY column does not have to be in the SELECT list.

```
SQL> SELECT      AVG(sal)
  2  FROM        emp
  3  GROUP BY deptno;
```

```
AVG (SAL)
-----
2916.6667
2175
1566.6667
```



Grouping by More Than One Column

EMP

DEPTNO	JOB	SAL
10	MANAGER	2450
10	PRESIDENT	5000
10	CLERK	1300
20	CLERK	800
20	CLERK	1100
20	ANALYST	3000
20	ANALYST	3000
20	MANAGER	2975
30	SALESMAN	1600
30	MANAGER	2850
30	SALESMAN	1250
30	CLERK	950
30	SALESMAN	1500
30	SALESMAN	1250

“sum salaries in the EMP table for each job, grouped by department”

DEPTNO	JOB	SUM (SAL)
10	CLERK	1300
10	MANAGER	2450
10	PRESIDENT	5000
20	ANALYST	6000
20	CLERK	1900
20	MANAGER	2975
30	CLERK	950
30	MANAGER	2850
30	SALESMAN	5600



Using the GROUP BY Clause on Multiple Columns

```
SQL> SELECT    deptno, job, sum(sal)
  2  FROM      emp
  3  GROUP BY  deptno, job;
```

DEPTNO	JOB	SUM(SAL)
-----	-----	-----
10	CLERK	1300
10	MANAGER	2450
10	PRESIDENT	5000
20	ANALYST	6000
20	CLERK	1900
...		

9 rows selected.





Illegal Queries

Using Group Functions

- Any column or expression in the SELECT list that is not an aggregate function must be in the GROUP BY clause.

```
SQL> SELECT deptno, COUNT(ename)
      2 FROM emp;
```

Column missing in the GROUP BY clause

```
SELECT deptno, COUNT(ename)
      *
```

```
ERROR at line 1:
```

```
ORA-00937: not a single-group group function
```





Illegal Queries

Using Group Functions

- You cannot use the WHERE clause to restrict groups.
- You use the HAVING clause to restrict groups.

```
SQL> SELECT      deptno, AVG(sal)
  2  FROM    emp
  3  WHERE      AVG(sal) > 2000
  4  GROUP BY  deptno;
```

```
WHERE AVG(sal) > 2000
```

```
*
```

```
ERROR at line 3
```

```
ORA-00934: group function is not allowed here
```

**Cannot use the WHERE clause
to restrict groups**



Excluding Group Results

EMP

DEPTNO	SAL
-----	-----
10	2450
10	5000
10	1300
20	800
20	1100
20	3000
20	3000
20	2975
30	1600
30	2850
30	1250
30	950
30	1500
30	1250

5000

3000

2850

“maximum
salary
per department
greater than
\$2900”

DEPTNO	MAX (SAL)
-----	-----
10	5000
20	3000



Example

- Employee (EID, Ename, DesignationID)
- Salary (DesgnID, Salary)
- Display ename, the highest, lowest, sum and average salary of all employees





Examples

- Employee (EID, Ename, DesignationID)
- Design (DesigID, Dname, Grade, SalID)
- Salary (SalID, Basic_salary, House_rent, Travelallowance, other allowances)
- How much is the average salary of designation (assistant professor in PUCIT)





Examples

- Department (DeptID, Salary)
- Calculate the maximum salary of each department





Excluding Group Results: HAVING Clause

- Use the HAVING clause to restrict groups
 - Rows are grouped.
 - The group function is applied.
 - Groups matching the HAVING clause are displayed.

```
SELECT column, group_function
FROM      table
[WHERE condition]
[GROUP BY group_by_expression]
[HAVING group_condition]
[ORDER BY column];
```





Using the HAVING Clause

```
SQL> SELECT      deptno, max(sal)
  2  FROM          emp
  3  GROUP BY      deptno
  4  HAVING        max(sal)>2900;
```

DEPTNO	MAX (SAL)
-----	-----
10	5000
20	3000





Having Clause

```
SELECT column_name, aggregate_function(column_name)
FROM table_name
WHERE column_name operator value
GROUP BY column_name
HAVING aggregate_function(column_name) operator value;
```





Using the HAVING Clause

```
SQL> SELECT      job, SUM(sal) PAYROLL
2  FROM          emp
3  WHERE         job NOT LIKE 'SALES%'
4  GROUP BY     job
5  HAVING        SUM(sal)>5000
6  ORDER BY     SUM(sal) ;
```

JOB	PAYROLL
ANALYST	6000
MANAGER	8275





Nesting Group Functions

- Display the maximum average salary.

```
SQL> SELECT      max (avg (sal) )  
      2  FROM      emp  
      3  GROUP BY deptno;
```

```
MAX (AVG (SAL) )  
-----  
      2916.6667
```





Summary

```
SELECT column, group_function(column)
FROM      table
[WHERE condition]
[GROUP BY group_by_expression]
[HAVING   group_condition]
[ORDER BY column];
```

- Order of evaluation of the clauses:
 - WHERE clause
 - GROUP BY clause
 - HAVING clause





Practice Overview

- Showing different queries that use group functions
- Grouping by rows to achieve more than one result
- Excluding groups by using the HAVING clause



EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

ASG

ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000

PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Display the name of employee who are elect engineers



EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

ASG

ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000

PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Display the name of employee who earn 25000



EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

ASG

ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

PNO	PNAME	BUDGET
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P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000

PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Display Eno, ename of employees who work on CAD/CAM Project and earn 25000



EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

ASG

ENO	PNO	RESP	DUR
E1	P1	Manager	12
E2	P1	Analyst	24
E2	P2	Analyst	6
E3	P3	Consultant	10
E3	P4	Engineer	48
E4	P2	Programmer	18
E5	P2	Manager	24
E6	P4	Manager	48
E7	P3	Engineer	36
E7	P5	Engineer	23
E8	P3	Manager	40

PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
P2	Database Develop.	135000
P3	CAD/CAM	250000
P4	Maintenance	310000

PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Count the number of employees on project CAD/CAM



EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

ASG

ENO	PNO	RESP	DUR
E1	P1	Manager	12
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PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Count the number of employees on each project name



EMP

ENO	ENAME	TITLE
E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

ASG

ENO	PNO	RESP	DUR
E1	P1	Manager	12
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E2	P2	Analyst	6
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E7	P3	Engineer	36
E7	P5	Engineer	23
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PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
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PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Display the name of employee having eno
E1, E2, E3, E4.

Display the name of employees working on
project

Instrumentation, maintenance or CAD/CAM



EMP

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E1	J. Doe	Elect. Eng.
E2	M. Smith	Syst. Anal.
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E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
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ASG

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PROJ

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PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Display the project names and eno who are one of the two P1, P3, OR the employee title is programmer . The query should also list the employees whose salary is greater than Salary of system analyst and mechanical engineer



EMP

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E2	M. Smith	Syst. Anal.
E3	A. Lee	Mech. Eng.
E4	J. Miller	Programmer
E5	B. Casey	Syst. Anal.
E6	L. Chu	Elect. Eng.
E7	R. Davis	Mech. Eng.
E8	J. Jones	Syst. Anal.

ASG

ENO	PNO	RESP	DUR
E1	P1	Manager	12
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PROJ

PNO	PNAME	BUDGET
P1	Instrumentation	150000
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PAY

TITLE	SAL
Elect. Eng.	40000
Syst. Anal.	34000
Mech. Eng.	27000
Programmer	24000

Display the name of employee who are elect engineers

